Annex III. Public Debt Sustainability Analysis

Korea Public Sector Debt Sustainability Analysis (DSA) - Baseline Scenario

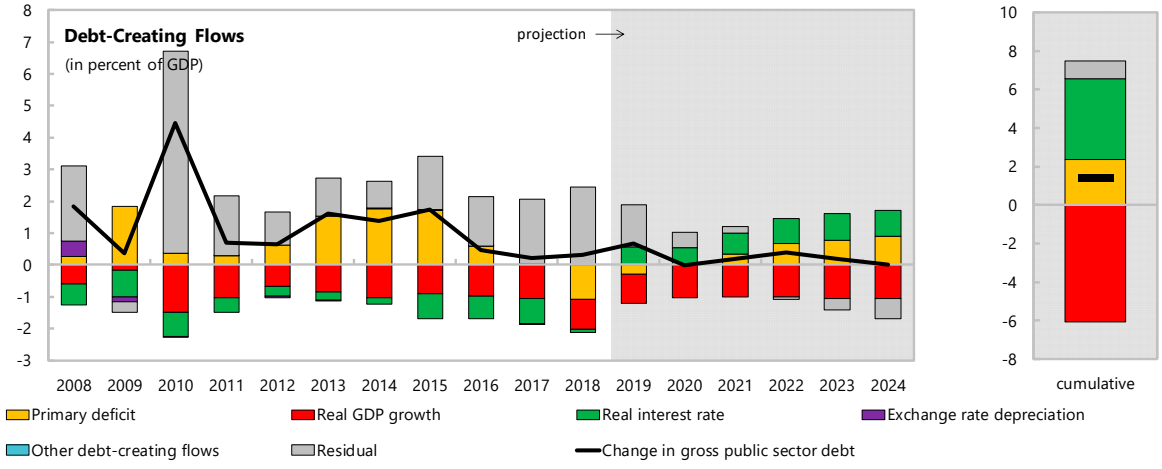
(in percent of GDP unless otherwise indicated)

Debt, Economic and Market Indicators [1/]

	Actual			Projections						As of March 29, 2019		
	2008-2016 [2/]	2017	2018	2019	2020	2021	2022	2023	2024	Sovereign Spreads		
Nominal gross central government debt	30.9	36.3	36.6	37.3	37.2	37.4	37.8	38.0	38.0	EMBIG (bp) 3/		-60
Gross financing needs	5.8	4.3	4.4	3.5	3.9	5.2	5.7	5.3	5.9	5Y CDS (bp)		32
Real GDP growth (in percent)	3.1	3.1	2.7	2.6	2.8	2.9	2.8	2.9	2.9	Ratings	Foreign	Local
Inflation (GDP deflator, in percent)	2.0	2.3	0.3	1.4	1.7	1.7	1.8	2.0	2.2	Moody's	Aa2	Aa2
Nominal GDP growth (in percent)	5.2	5.4	3.0	4.0	4.5	4.6	4.7	4.9	5.2	S&Ps	AA	AA
Effective interest rate (in percent) [4/]	3.8	2.4	3.0	3.0	3.2	3.6	4.0	4.3	4.5	Fitch	AA-	AA-

Contribution to Changes in Public Debt

	Actual			Projections							
	2008-2016	2017	2018	2019	2020	2021	2022	2023	2024	cumulative	debt-stabilizing primary balance [9/]
Change in gross central government debt	1.5	0.2	0.3	0.7	0.0	0.2	0.4	0.2	0.0	1.4	
Identified debt-creating flows	-0.4	-1.8	-2.1	-0.6	-0.5	0.0	0.4	0.5	0.7	0.5	-0.2
Primary deficit (excluding Social Security Fund)	1.0	0.0	-1.1	-0.3	0.0	0.3	0.7	0.8	0.9	2.4	
Primary (noninterest) revenue and grants	16.8	17.8	18.8	19.2	19.3	19.2	19.1	19.1	19.1	115.1	
Primary (noninterest) expenditure	17.8	17.8	17.7	18.9	19.2	19.5	19.8	19.9	20.0	117.5	
Automatic debt dynamics [5/]	-1.4	-1.9	-1.0	-0.3	-0.5	-0.4	-0.2	-0.2	-0.2	-1.9	
Interest rate/growth differential [6/]	-1.4	-1.8	-1.1	-0.3	-0.5	-0.4	-0.2	-0.2	-0.2	-1.9	
Of which: real interest rate	-0.5	-0.8	-0.1	0.6	0.6	0.7	0.8	0.8	0.8	4.2	
Of which: real GDP growth	-0.9	-1.0	-0.9	-0.9	-1.0	-1.0	-1.0	-1.1	-1.1	-6.1	
Exchange rate depreciation [7/]	0.0	0.0	0.0	...	...	...	...	...	...	...	
Other identified debt-creating flows	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Net privatization proceeds (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Please specify (2) (e.g., ESM and Euroarea loar	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Residual, including asset changes [8/]	1.8	2.1	2.4	1.3	0.5	0.2	-0.1	-0.4	-0.6	0.9	



Source: IMF staff.

1/ Public sector is defined as central government.
Data refer to the central government excluding the Social Security Fund (SSF) as the central government balance excluding the SSF drives the debt dynamics. The SSF accounts are in surplus and the SSF is accumulating assets. Data on other parts of the general government are not included as they become available with a significant lag.

2/ Based on available data.

3/ Long-term bond spread over U.S. bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

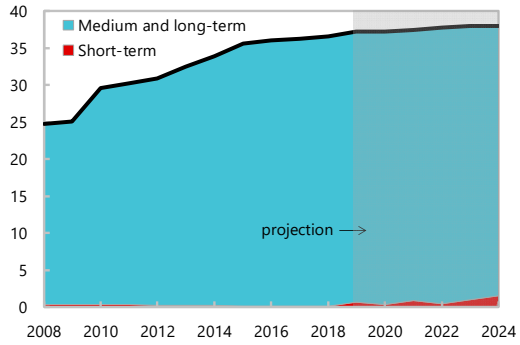
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Korea Public DSA - Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

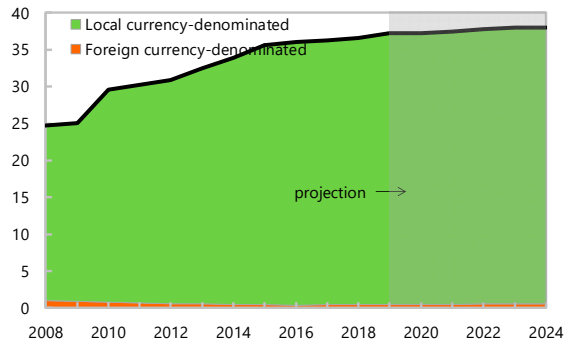
By Maturity

(in percent of GDP)

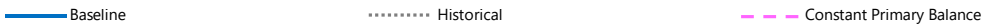


By Currency

(in percent of GDP)

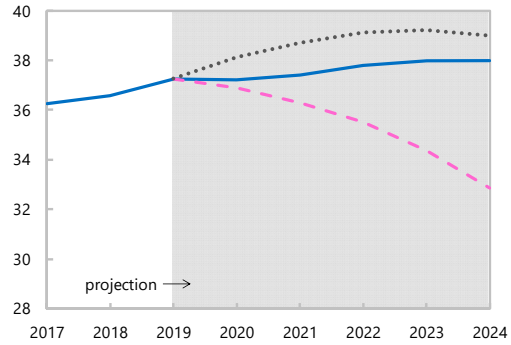


Alternative Scenarios



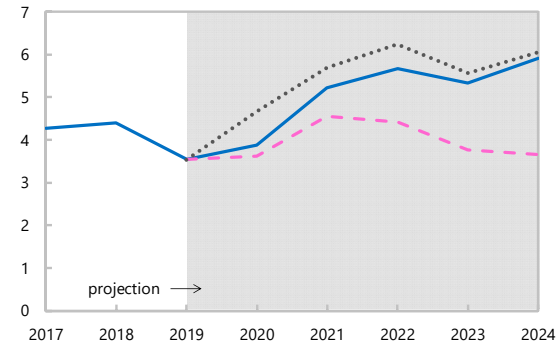
Gross Nominal Public Debt

(in percent of GDP)



Public Gross Financing Needs

(in percent of GDP)



Underlying Assumptions

(in percent)

Baseline Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	2.6	2.8	2.9	2.8	2.9	2.9
Inflation	1.4	1.7	1.7	1.8	2.0	2.2
Primary Balance	0.3	0.0	-0.3	-0.7	-0.8	-0.9
Effective interest rate	3.0	3.2	3.6	4.0	4.3	4.5
Constant Primary Balance Scenario						
Real GDP growth	2.6	2.8	2.9	2.8	2.9	2.9
Inflation	1.4	1.7	1.7	1.8	2.0	2.2
Primary Balance	0.3	0.3	0.3	0.3	0.3	0.3
Effective interest rate	3.0	3.2	3.6	4.0	4.3	4.5

Historical Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	2.6	3.1	3.1	3.1	3.1	3.1
Inflation	1.4	1.7	1.7	1.8	2.0	2.2
Primary Balance	0.3	-0.8	-0.8	-0.8	-0.8	-0.8
Effective interest rate	3.0	3.2	3.5	3.9	4.1	4.3

Source: IMF staff.

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